Group 3 :
Pollinators Detection

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Credits

- Instructors: Lisheng Sun-Hosoya (class design and coordination)
- Ihsan Ullah and Thanh Gia Hieu Khuong (TP)
- Class initiated by Prof. Isabelle Guyon

Project Overview

- Phase 1 (4 weeks): Design and deploy the Pollinators Detection Challenge
- Phase 2 (2 weeks): Participate in the Grain Classification Challenge

Phase 1: Challenge Definition

Purpose of the challenge:

- Binary Classification
- Predict insect presence or absence
- Lightweight model

Codabench

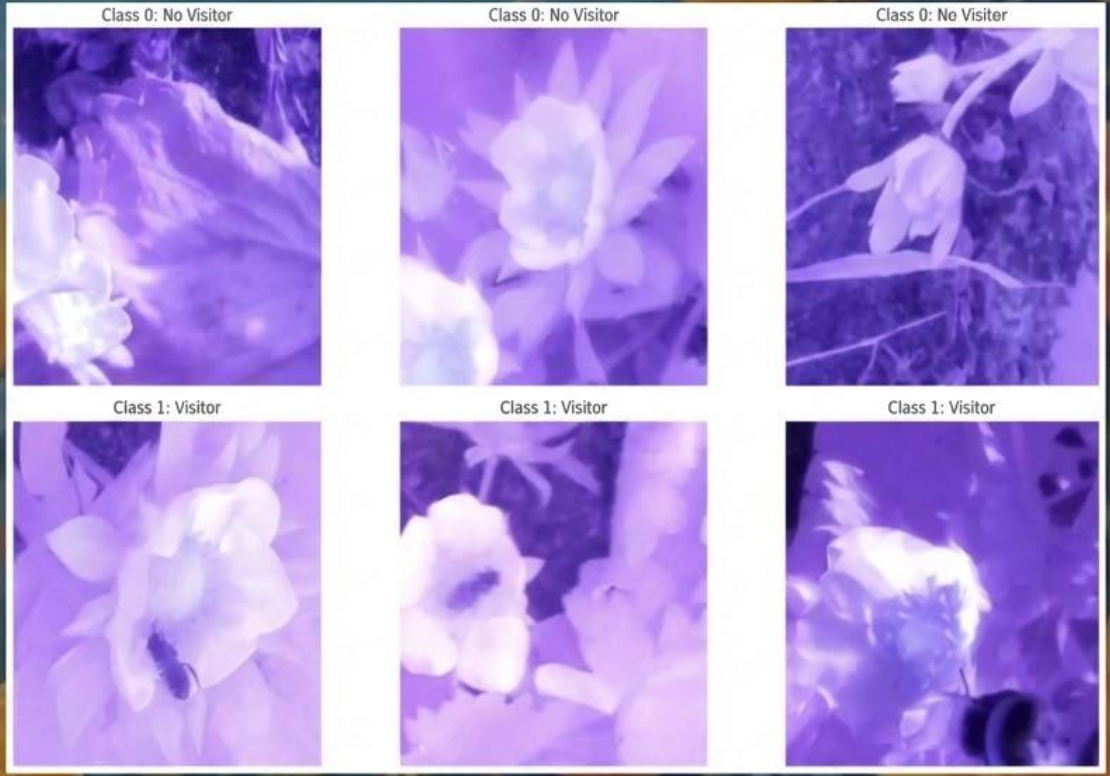
Data Science Competitions and Benchmarks

Phase 1: Dataset Design

Data from INRAE

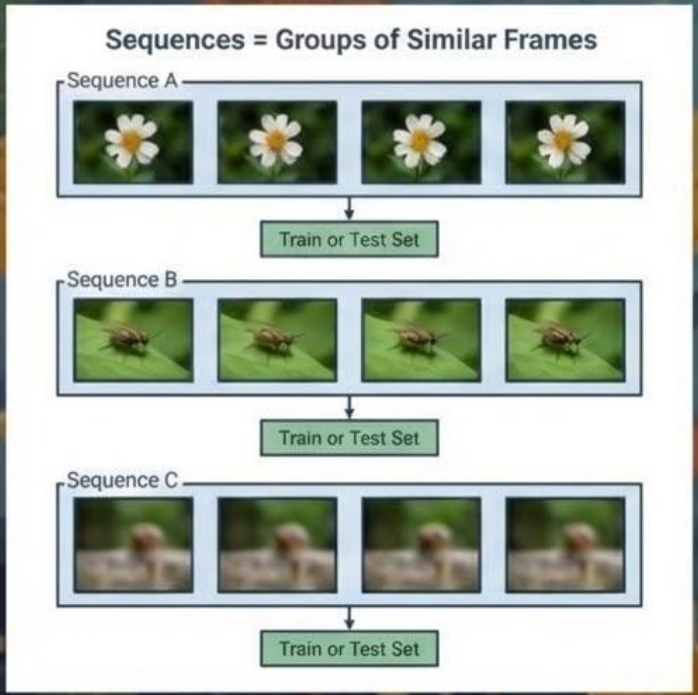
HDF5 and JPG files

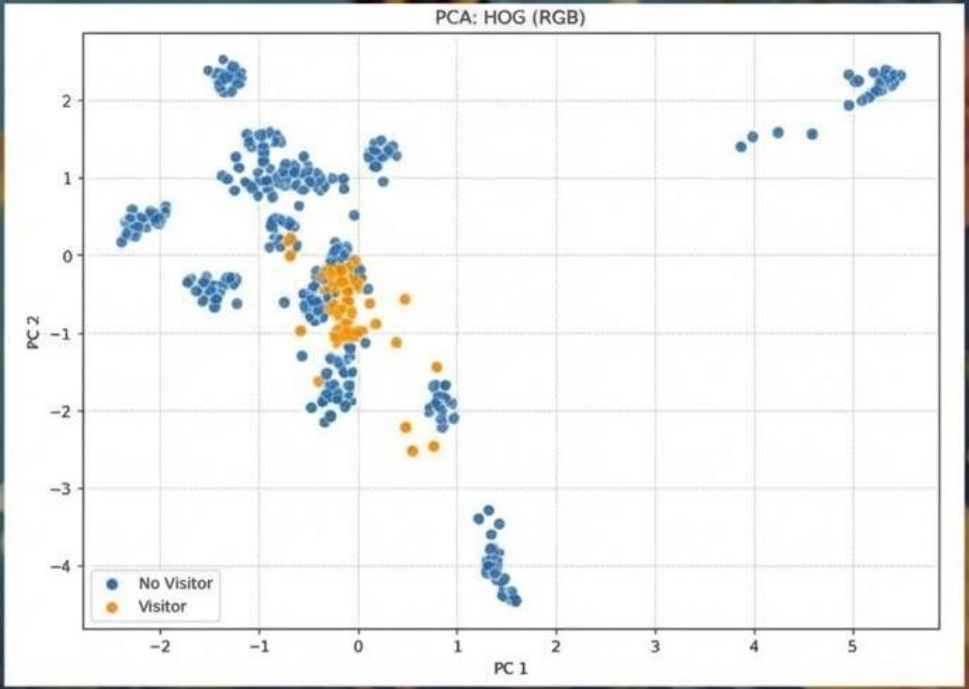
224x224, RGB channels



Phase 1: Dataset Design

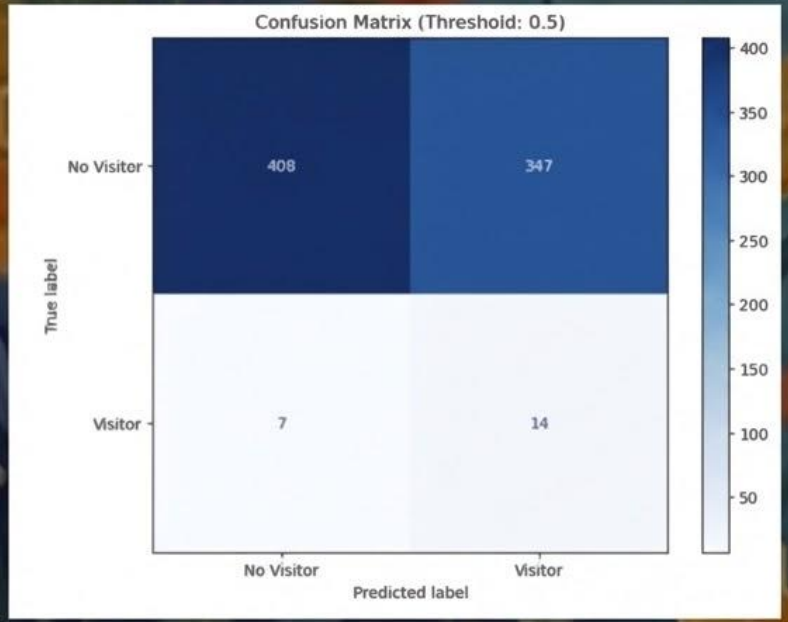
Even distribution between test and train + no data leakage





Phase 1: Target and Evaluation

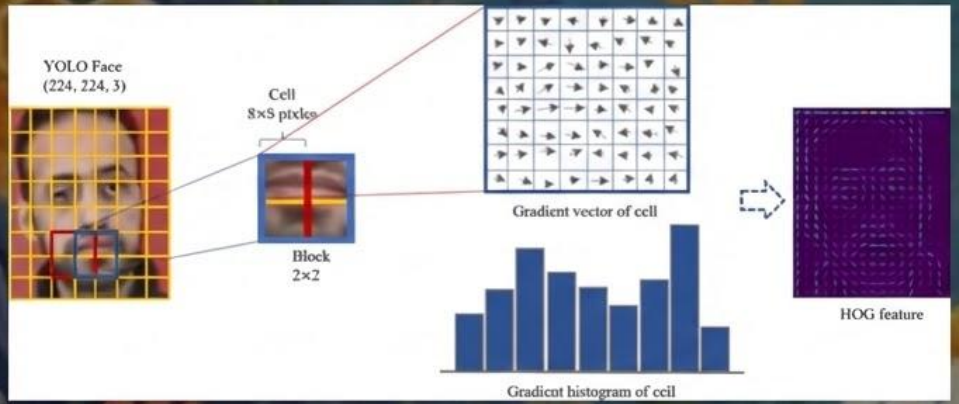
Can players detect insects ? Assessed with F1-Score and Balanced Accuracy

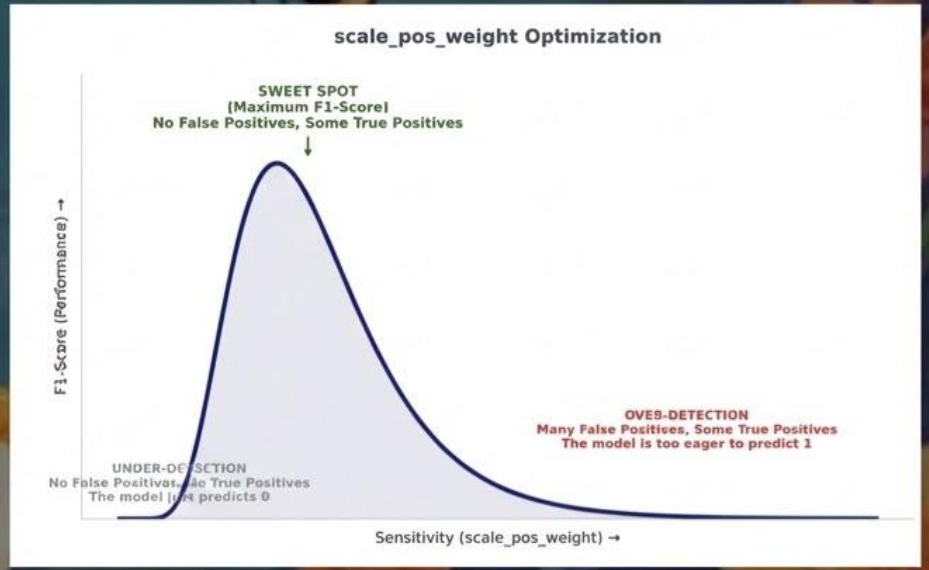


Leaderboard

Task:				Public Task	
#	Participant	Date	ID	F1 Score	Balanced Accuracy
1	baptistepras2nd	2026-01-31 17:45	514962	0.336	0.661
2	Ihsan Ullah	2026-02-07 19:23	529122	0.336	0.661
3	amoujar	2026-02-09 16:25	532683	0.336	0.661
4	olutola_paul	2026-02-09 19:21	533111	0.336	0.661

Phase 1: Baseline Methods



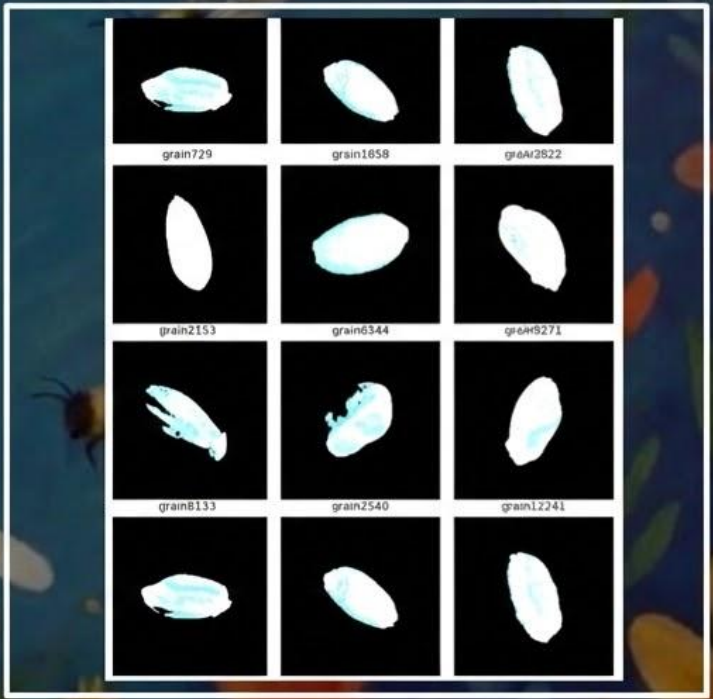


PHASE 2: Solving Grain Variety Classification Challenge

Task: Classification of each grain images into 8 varieties

Baseline: Handcrafted features + RandomForest (Provided by Group 2)

Objective: Improve classification accuracy by exploring different models



Phase 2: Chosen Challenge

The Addressed Problem

- Grain varieties are mixed after harvest
- Differences between varieties are subtle
- Reliable automatic identification is needed

The ML task

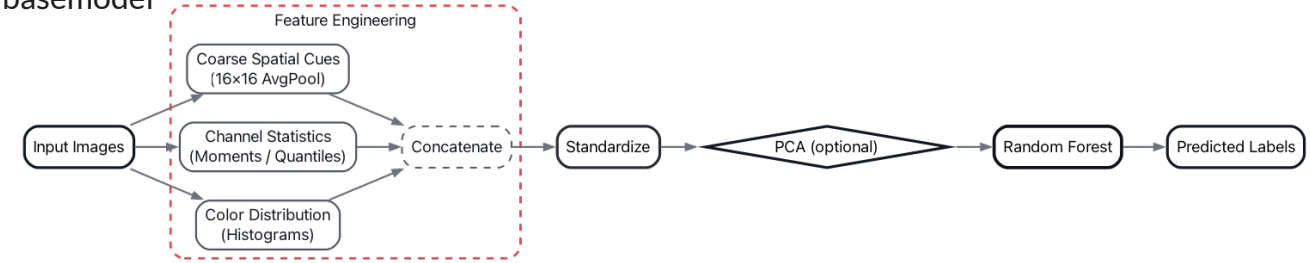
- Input: single grain 3 channel image
- Output: 1 of 8 varieties
- Task: multi-class classification
- Metric: accuracy

Why chosen?

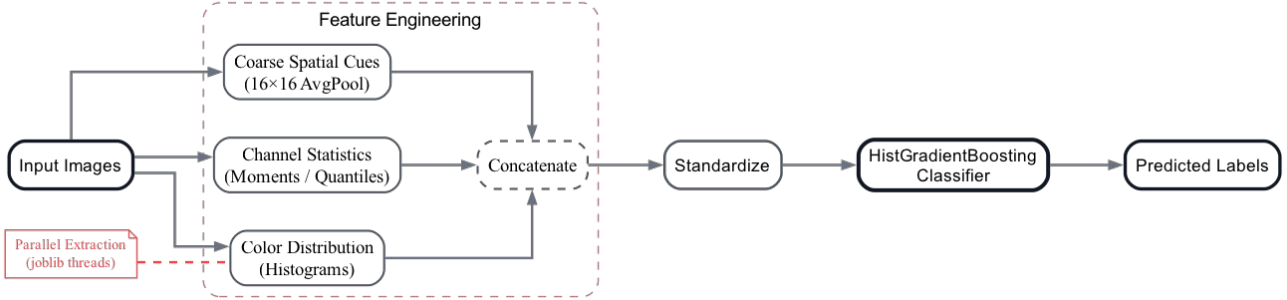
- ✓ Real agricultural application
- ✓ Classifying objects using machines when visual differences are minimal is challenging. It requires comparing the limits of machines with the human capabilities.
- ✓ Opportunity to compare multiple classification models

Phase 2: Submissions

basemodel



first approach



key differences & advantage

- Boosting instead of bagging → stronger modeling capacity
- No PCA → full feature information retained
- Parallel extraction → scalable preprocessing

Accuracy improved from 0.53 to 0.62

Improved approach

End-to-ened CNN (ResNet18)

- Input: 252×252×3
- Random crop to 224-22243
- Data augmetation
- Fully augnentation
- Fully connected → 8 classes

Training details

- AdamW optimizer
- Learning rate = 3e-4
  Weight decay = 1e-4
- Early stopping (patience=6)
- Validation split = 15%

Data augnentation

- Random flip
- Random rotation
  Random crop

Why ResNet?

- Deep architecture
- Residual connections
- Stable training
- Proven performance in image classification

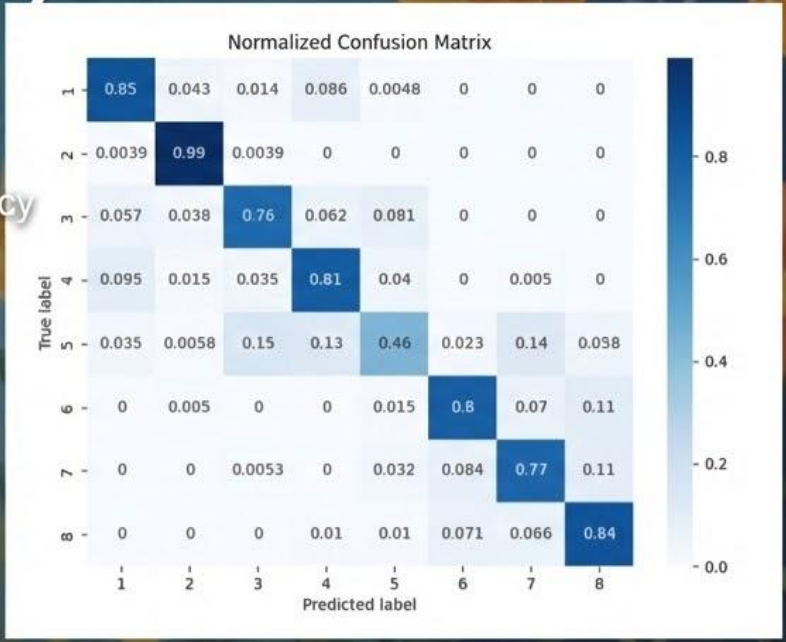
Approach	Feature Source	Learning Type
RF + PCA	Hand-crafted features	Bagging (ensemble)
HGB	Hand-crafted features	Boosting (ensemble)
Model1	**Learned automatically**	**Deep CNN**

Phase 2: Results and Analysis

Overall performance
Strong diagonal dominance → good classification accuracy
Classes 2, 1, 4, 6, 7, 8 achieve ~0.8–0.99 accuracy
Errors are not random

Challenging classes
Class 5: lowest accuracy (~0.46)
Frequently confused with classes 3, 4, and 7
Some confusion among classes 6, 7, and 8

Most errors occur between spectrally similar varieties
Similar texture and structure patterns
Limited 3-band information may restrict separability



Phase 2: Results and Analysis

Why does Model1 outerforom traditional ML?

- Learns spatial structure
- Learns complex feature interactions
- End-o-endd optimization

#	Participant	Date	ID	Score
1	baptistepras	2026-02-16 23:17	545545	0.81
2	charlottelee	2026-02-14 03:16	541259	0.8
3	charlottelee	2026-02-13 20:57	540824	0.71
4	baptistepras	2026-02-09 22:51	532573	0.62
5	baptistepras	2026-02-09 20:25	532231	0.56
6	olutola_paul	2026-01-26 03:19	504071	0.56

No information loss from manual features.
in fixed features.

CNN learns adaptive represtationns tailroed ed
the classification objective.

Error-based compuation and and analysis

Confusion analysis

- Most errors occur between spectrally similar similar varieties

Possible causes:

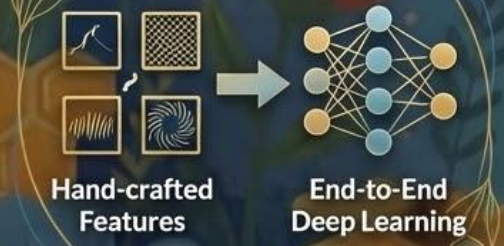
Similar texture

- Limited 3-band information

Future work:

- Use more spectral bands
- Larger CNN
- Transfer learning

WrapUp



Transitioned from hand-crafted features to end-to-end deep learning

CNN outperformed traditional ML by capturing texture and structure

Performance gain came from both architecture and training strategy

Conclusion: What I learned

As a challenge organizer

As a challenge participant

- Importance of data understanding (3-band vs full hyperspectral)
- Difference between feature engineering and representation learning
- Impact of model choice on structured vs image data
- Role of regularization and validation strategy
- Practical engineering constraints (environment, memory, compatibility)

Future work

If given more time / ressource, what are possible improvement in:

creation phase

resolution phase

- Try deeper architectures (ResNet34/50)
- Explore spectral attention mechanisms
- Cross-validation instead of single split
- Ensemble multiple CNN models

Ressources and Links

challenge website : M1 AI Challenge 2025-26 | Group 3 - Pollinators - Codabench

github repo : vladihnat/AI-Challenge-2025-26-Group-3-: Master 1 AI challenge creation and resolution